Data Analytics Capstone

Accuracy and Fairness Trade-offs in Machine Learning Credit Default Prediction: A Threshold-Level Governance Study for Retail Lenders

Synopsis

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Introduction

Background and Context

The credit decisions I review for Indian banks and non-banking financial companies (NBFCs) are now produced, in most new retail programmes, by gradient-boosted tree models rather than by logistic scorecards. Ensemble methods have improved on logistic regression in most published benchmarks, although the gain varies by dataset: Lessmann et al. (2015) compared 41 classifiers on eight credit datasets and found boosted and bagged trees ahead on the area under the receiver operating characteristic curve (AUC) in most cases. What is less often measured is whether the model treats groups of borrowers differently and by how much. In my experience that question is answered with a policy statement, usually that sex and marital status are excluded from the feature set, not with a number. Excluding the protected attribute does not establish that the resulting decisions are free of group-level differences, because other features can carry the same information (Kamiran & Calders, 2012).

Regulatory attention to fairness in financial AI has increased, although the specific obligations depend on jurisdiction and instrument. The Reserve Bank of India (RBI) FREE-AI Committee report of 13 August 2025 sets out seven guiding principles and 26 recommendations, among them disclosure of AI-driven credit assessments and fairness provisions proposed for the Digital Lending Directions. The recommendations remain advisory until the RBI adopts them (Reserve Bank of India, 2025). The European Union's Artificial Intelligence Act lists credit scoring of natural persons as a high-risk use in Annex III (European Parliament & Council of the European Union, 2024), a point of comparison rather than applicable Indian law. The United States literature has measured the problem: Fuster et al. (2022) found that machine learning in mortgage underwriting raised accuracy while widening the dispersion of predicted default across racial groups, and Bartlett et al. (2022) found algorithmic lenders charging Latinx and African-American borrowers higher rates.

The measurement tools exist. Hardt et al. (2016) defined equality of opportunity as equal true-positive rates for the favourable outcome across groups, Feldman et al. (2015) expressed the

four-fifths rule from United States employment-selection guidelines (Uniform Guidelines on Employee Selection Procedures, 1978) as a disparate impact ratio, and Kleinberg et al. (2017) proved that calibration and equal error rates cannot both hold when base rates differ, so a lender must choose which criterion to govern to. Kozodoi et al. (2022) applied these criteria to seven credit datasets, found that fairness could be reached at a low profit cost, and identified separation, the family that includes equal opportunity, as the suitable criterion for a scorecard. This study places the trade-off in a threshold-level decision table using the measures validation teams already read: retention rate, default rate among retained customers, AUC, and the fairness gap at each cut-off. My position, developed in the AI strategy coursework of this programme, is that fairness in a deployed model is decided by configuration choices such as the threshold and the sample weights rather than by governance documents, and this capstone tests that position.

Problem Statement

Retail lenders deploy machine learning default models that rank borrowers more accurately than logistic scorecards, but they lack a quantified, reproducible measure of how those models' decisions differ by demographic group and of what it costs to narrow the difference. Validation reports record the AUC and the Kolmogorov–Smirnov (KS) statistic and treat fairness as satisfied once protected attributes are dropped, an approach called fairness through unawareness that fails when proxy variables remain (Mehrabi et al., 2021). The gap this project addresses is the absence of a trade-off curve, estimated on an open credit dataset with stated statistical power, that expresses the cost of two standard mitigation methods in the units of a credit policy and ends in a decision rule for the threshold. The setting is a retrospective portfolio-retention simulation on existing cardholders: the dataset records repayment outcomes for customers who already held credit, so a threshold applied to it simulates which cardholders a lender would have retained or referred, not whom it would have approved. The dependent variable is DEFAULT, next-month payment default, coded 1 or 0. The independent variables are the 23 borrower attributes in the dataset, listed in Table 2. Two derived decision variables, FLAG (simulated referral or decline) and RETAIN (simulated favourable decision), carry the fairness

audit. Sex is the primary protected attribute and age band the secondary one. The problem is a supervised binary classification task with an attached fairness audit.

Purpose of the Study

The study has four aims, one per research question. First, it predicts: it classifies cardholders as likely or unlikely to default next month using three model families and tests a pre-specified contrast between gradient boosting and the logistic baseline. Second, it tests: it estimates how far the observed default rate differs by sex and by age band, which decides which fairness criteria can hold together (Kleinberg et al., 2017). Third, it audits: it measures, at a simulated cut-off, how favourable-selection rates and false-decline rates differ by sex and compares the differences with predefined research benchmarks. Fourth, it quantifies: it applies two mitigation methods and records the predictive-performance or decision-quality cost of each. The deliverable is a reproducible validation workflow and an auditable decision-support table, not an algorithm that certifies fairness.

Scope and Objectives

Research Questions

Research Question 1 (RQ1): Predictive Performance

How do logistic regression, random forest (Breiman, 2001), and gradient boosting (Chen & Guestrin, 2016) compare in predicting next-month default, measured by AUC with a 95% confidence interval (CI), and is the AUC improvement of the pre-specified gradient boosting model over the logistic baseline statistically significant under the DeLong et al. (1988) test?

$H1_0$: $AUC_{GB} = AUC_{LR}$. $H1_1$: $AUC_{GB} > AUC_{LR}$. Gradient boosting against logistic regression is the single pre-specified contrast. Random forest is a secondary benchmark and is not tested for significance. The dependent variable is DEFAULT and the independent variables are the 23 attributes in Table 2.

Research Question 2 (RQ2): Outcome Disparity

Do observed default rates differ by sex and by age band, and how large are the differences before and after adjustment for credit limit and age?

$H2_0$: $p_{male} = p_{female}$. $H2_1$: $p_{male} \neq p_{female}$. For age bands, $H2b_0$ is independence of default and age band. A difference of 0.03 is the planned minimum meaningful difference used for power, not part of the hypothesis. The dependent variable is DEFAULT and the independent variables are SEX and the derived AGE_BAND. Observed differences are not attributed causally to sex or age.

Research Question 3 (RQ3): Fairness at a Simulated Cut-off

At a simulated cut-off that retains 75% of existing cardholders, how do favourable-selection rates and false-decline rates differ by sex, and how do those differences compare with two predefined research benchmarks, a disparate impact ratio of at least 0.80 and a false-decline rate gap within ±0.05?

$H3_0$: $FDR_{male} = FDR_{female}$, where FDR is the false-decline rate, the share of non-defaulters flagged. $H3_1$: $FDR_{male} \neq FDR_{female}$. The benchmark comparison is descriptive: the study reports whether the 95% CI for the gap lies within ±0.05 and whether the disparate impact (DI) ratio on RETAIN rates reaches 0.80. The true-positive rate gap among defaulters is a secondary, lender-risk measure. The dependent variables are the RETAIN rate and the false-decline rate per group and the independent variable is SEX.

Research Question 4 (RQ4): Mitigation Trade-offs

How do training-data reweighing (Kamiran & Calders, 2012) and group-specific threshold adjustment (Hardt et al., 2016) change the false-decline rate gap and the simulated portfolio outcomes, and what predictive-performance or decision-quality cost arises under each?

The two methods are judged on different measures because they act at different points. Reweighing retrains the model and can change the ranking, so $H4_0$: $AUC_{RW} \leq AUC_{base} - 0.01$ and $H4_1$: $AUC_{RW} > AUC_{base} - 0.01$, a one-sided non-inferiority test on paired AUCs. Threshold

adjustment leaves the scores untouched, so its AUC is unchanged by construction and its cost is the default rate among retained cardholders at the common 75% retention rate, with a bootstrap CI, against the unmitigated cut-off.

Scope Boundaries

In scope are one public dataset of 30,000 cardholders, three model families, one primary protected attribute (sex) and one secondary attribute (age band), two fairness measures used as tests (the DI ratio on RETAIN and the false-decline rate gap) with the true-positive rate gap and equalized odds reported alongside, two mitigation methods, and a trade-off table across simulated retention rates. Shapley additive explanations (SHAP) (Lundberg & Lee, 2017) to identify candidate proxy features are optional and will be dropped first if the schedule tightens. Out of scope are neural network classifiers, whose interpretability cost on tabular high-stakes decisions is not repaid by their accuracy gain (Rudin, 2019), synthetic data, in-processing mitigation, calibration-based criteria, causal inference, profit optimisation, reject inference, and deployment.

Feasibility and Measurability

The dataset downloads from the UCI Machine Learning Repository without registration under a Creative Commons Attribution 4.0 licence (Yeh, 2009). Every variable is numeric or integer-coded, and every outcome in RQ1 to RQ4 is a proportion, an AUC, or a difference between two of them, so all variables are measurable without transformation. A 30,000-row file with 23 features is small enough that nested tuning and bootstrapping take minutes to an hour on a laptop, and the exact runtime is an estimate until benchmarked in week 2. The tools are Python 3.11 with pandas, scikit-learn, XGBoost, Fairlearn (Bird et al., 2020), SHAP, and statsmodels. Table 4 sets out the phased plan.

Sample Size Calculation

All four research questions are evaluated on pooled out-of-fold predictions and decisions from stratified five-fold cross-validation (CV). Each outer training set is split into a fit partition (80%), on which models are trained and tuned by inner CV, and a calibration partition (20%), on

which the retention cut-off, the group thresholds, and the reweighing weights are set. The held-out outer fold is scored once with the fitted model and the calibrated policy and never used to choose anything, so each cardholder receives one prediction and one decision from a configuration that did not see that cardholder, and the evaluation pool equals the full 30,000. The calculations use $\alpha = 0.05$, power $1 - \beta = 0.80$, $z_{0.975} = 1.960$, $z_{0.95} = 1.645$, and $z_{0.80} = 0.842$ throughout. The overall default prevalence is 22.12% (6,636 of 30,000). From my initial tabulation of the file, 39.63% of cardholders are male (11,888) and 60.37% female (18,112), with 2,873 and 3,763 defaulters respectively. These counts enter RQ2 to RQ4 as allocation ratios.

RQ1: Confidence Interval for the AUC and Power for the Paired Contrast

RQ1 commits to two things: reporting each model's AUC to a stated precision, and testing one pre-specified paired contrast. The first uses the confidence interval approach of Hanley and McNeil (1982), which gives the standard error of an AUC A from the numbers of defaulters n_D and non-defaulters n_N, where $Q_1 = A/(2 - A)$ and $Q_2 = 2A^2/(1 + A)$.

$$SE(A) = \sqrt{\frac{A(1-A) + (n_D - 1)(Q_1 - A^2) + (n_N - 1)(Q_2 - A^2)}{n_D n_N}} \quad (1)$$

Step 1 sets the planning AUC at $A = 0.78$, the value I expect for tuned gradient boosting from prior work with similar card portfolios. Step 2 computes the constants: $Q_1 = 0.78/1.22 = 0.6393$, $Q_2 = 1.2168/1.78 = 0.6836$, and $A^2 = 0.6084$, so $Q_1 - A^2 = 0.0309$ and $Q_2 - A^2 = 0.0752$. Step 3 substitutes the prevalence. With $n_D = 0.2212n$ and $n_N = 0.7788n$, and n large enough that the minus-one terms are immaterial, Equation 1 reduces to $SE^2 = [0.2212(0.0309) + 0.7788(0.0752)] / (0.2212 \times 0.7788 \times n) = 0.0654/(0.1723n) = 0.3797/n$. Step 4 sets the precision target: a 95% CI half-width of 0.01 requires $SE \leq 0.01/1.960 = 0.005102$, so $SE^2 \leq 2.603 \times 10^{-5}$. Step 5 solves for n: $n \geq 0.3797/(2.603 \times 10^{-5}) = 14{,}587$, and the exact solution of Equation 1 gives the same figure. At $A = 0.75$ the constant rises to 0.4114 and the requirement to 15,805.

The second calculation powers the DeLong test of H1. Two AUCs estimated on the same cardholders are correlated, and with equal standard errors $\text{Var}(A_{GB} - A_{LR}) = 2(1 - r)SE^2$, where r

is the correlation between the two estimates. Step 6 sets the minimum detectable improvement at $\Delta = 0.02$, below the four to six points I expect, with two-sided $\alpha = 0.05$ and power 0.80. Step 7 sets $r = 0.70$, the assumption discussed under RQ4. Step 8 solves $n \geq 2(1 - r) \times 0.3797 \times (z_{0.975} + z_{0.80})^2/\Delta^2 = 0.6 \times 0.3797 \times 7.849/0.0004 = 4{,}470$, or 7,450 at $r = 0.50$. The paired test is powered at any sample that meets the precision target, so RQ1 takes 14,587 as its minimum.

RQ2: Power Analysis for Two Proportions

RQ2 compares two independent proportions with unequal group sizes, so I use the power formula for a two-sided two-proportion z test with allocation ratio k (Fleiss et al., 2003), where n_1 is the smaller group (male), $n_2 = kn_1$ is the larger (female), $\bar{p} = (p_1 + kp_2)/(1 + k)$, and $\Delta = p_1 - p_2$.

$$n_1 = \frac{\left[z_{1-\alpha/2} \sqrt{(1+1/k)\,\bar{p}(1-\bar{p})} + z_{1-\beta} \sqrt{p_1(1-p_1) + p_2(1-p_2)/k} \right]^2}{\Delta^2} \tag{2}$$

Step 1 sets the planned minimum meaningful difference at $\Delta = 0.03$, with planning values $p_1 = 0.24$ and $p_2 = 0.21$. A three-point gap is the smallest I treat as material because it moves a segment across the pricing bands most card issuers use. Step 2 sets $k = 18{,}112/11{,}888 = 1.524$. Step 3 computes $\bar{p} = (0.24 + 1.524 \times 0.21)/2.524 = 0.2219$ and $1 - \bar{p} = 0.7781$. Step 4 evaluates the two terms in the bracket. The first is $1.960 \times \sqrt{[(1 + 1/1.524) \times 0.2219 \times 0.7781]} = 1.960 \times \sqrt{(1.656 \times 0.1727)} = 1.960 \times \sqrt{0.2860} = 1.960 \times 0.5348 = 1.048$. The second is $0.842 \times \sqrt{[0.24 \times 0.76 + 0.21 \times 0.79/1.524]} = 0.842 \times \sqrt{(0.1824 + 0.1089)} = 0.842 \times \sqrt{0.2913} = 0.842 \times 0.5397 = 0.454$. Step 5 squares and divides: $n_1 = (1.048 + 0.454)^2/0.03^2 = 2.257/0.0009 = 2{,}508$ male cardholders, and $n_2 = 1.524 \times 2{,}508 = 3{,}821$ female cardholders, 6,329 in total. Step 6 converts to a dataset size: $N = 2{,}508/0.3963 = 6{,}329$, since the sexes are drawn in the file's own proportions. Cohen's (1988) h for this difference is $2 \arcsin\sqrt{0.24} - 2 \arcsin\sqrt{0.21} = 0.072$, a small effect. The age-band test is a secondary 2 × 4 chi-square with three degrees of freedom. For a small effect $w = 0.10$ at $\alpha = 0.05$ and power 0.80 the non-centrality parameter is 10.91 (Cohen, 1988), so $N = 10.91/0.10^2 = 1{,}091$, which the sex comparison dominates.

RQ3: Power Analysis for the False-Decline Rate Gap

RQ3 tests a difference in false-decline rates, which only non-defaulters enter, so the comparison groups are the 9,015 male and 14,349 female non-defaulters, and Equation 2 applies with the counts restricted to non-defaulters and then inflated to the full population. Step 1 sets the minimum detectable gap at $\Delta = 0.05$, the benchmark in RQ3, with planning values $FDR_{male} = 0.19$ and $FDR_{female} = 0.14$, bracketing the overall rate of about 0.165 implied by a 25% flag rate at an AUC near 0.78. Step 2 sets $k = 14{,}349/9{,}015 = 1.592$. Step 3 computes $\bar{p} = (0.19 + 1.592 \times 0.14)/2.592 = 0.1593$ and $1 - \bar{p} = 0.8407$. Step 4 evaluates the bracket. The first term is $1.960 \times \sqrt{[(1 + 1/1.592) \times 0.1593 \times 0.8407]} = 1.960 \times \sqrt{(1.628 \times 0.1339)} = 1.960 \times \sqrt{0.2181} = 1.960 \times 0.4670 = 0.915$. The second is $0.842 \times \sqrt{[0.19 \times 0.81 + 0.14 \times 0.86/1.592]} = 0.842 \times \sqrt{(0.1539 + 0.0756)} = 0.842 \times \sqrt{0.2295} = 0.842 \times 0.4791 = 0.403$. Step 5 squares and divides: $(0.915 + 0.403)^2/0.05^2 = 1.738/0.0025 = 695$ male non-defaulters, and $1.592 \times 695 = 1{,}107$ female non-defaulters, 1,802 in total. Step 6 inflates to the full population. Male non-defaulters are $0.3963 \times 0.7583 = 0.3005$ of all cardholders, so $N = 695/0.3005 = 2{,}314$. At the full 30,000 the 95% CI for the gap has a half-width of about $1.960 \times \sqrt{[0.165 \times 0.835 \times (1/9{,}015 + 1/14{,}349)]} = 0.010$, so a point estimate within ±0.04 places the whole interval inside the ±0.05 band.

RQ4: Non-Inferiority Test for Paired AUCs After Reweighing

RQ4's inferential test applies to reweighing only, because threshold adjustment does not change the scores. The baseline and reweighed models score the same cardholders, so their AUCs are correlated, and the appropriate comparison is the correlated-AUC test of DeLong et al. (1988) framed as a one-sided non-inferiority test. Step 1 sets the margin at $\delta = 0.01$, the largest AUC loss the lender accepts in exchange for closing the fairness gap, with one-sided $\alpha = 0.05$, power 0.80, and a true difference of zero. Step 2 sets $r = 0.70$. I have not found a published estimate of the AUC correlation between a model and its reweighed retraining, so 0.70 is a planning assumption, chosen below the correlation two models fitted to the same features on the same rows would usually show, and checked in Step 6. Step 3 sets the requirement $\delta/SE_{diff} \geq z_{0.95} + z_{0.80} = 1.645 + 0.842 = 2.487$, so $SE_{diff} \leq 0.01/2.487 = 0.00402$. Step 4 converts to the single-

AUC standard error: $SE_{diff} = \sqrt{(2 \times 0.30)} \times SE = 0.7746 \times SE$, so $SE \leq 0.00402/0.7746 = 0.00519$. Step 5 applies the RQ1 variance constant: $n \geq 0.3797/0.00519^2 = 0.3797/(2.694 \times 10^{-5}) = 14{,}094$, and the exact solution of Equation 1 gives 14,086. Step 6 checks sensitivity: at $r = 0.85$ the requirement falls to 7,042, and at $r = 0.50$ it rises to 23,474, still inside the available 30,000. The decision-level measures for threshold adjustment carry 1,000-sample stratified bootstrap intervals rather than a power calculation.

Table 1

Sample Size Requirement by Research Question

Research question	Method used	Key parameters	Minimum sample size (N)
RQ1 model performance	Confidence interval for AUC (Hanley & McNeil, 1982), plus power for the paired DeLong contrast	A = 0.78, prevalence = 0.2212, 95% CI half-width e = 0.01, z = 1.960; paired: Δ = 0.02, r = 0.70, α = 0.05 (two-sided), power = 0.80	14,587 (precision); 4,470 (paired test)
RQ2 outcome disparity	Power analysis, two-proportion z test, unequal allocation (Fleiss et al., 2003)	α = 0.05 (two-sided), power = 0.80, p1 = 0.24, p2 = 0.21, Δ = 0.03, k = 1.524, Cohen's h = 0.072	6,329 (2,508 male, 3,821 female)
RQ3 fairness at cut-off	Power analysis, two-proportion z test on false-decline rate within non-defaulters, inflated by non-default share	α = 0.05, power = 0.80, FDR = 0.19 vs 0.14, Δ = 0.05, k = 1.592, male non-defaulter share = 0.3005	2,314 (1,802 non-defaulters)
RQ4 mitigation trade-offs	Power analysis, one-sided non-inferiority of correlated AUCs (DeLong et al., 1988), reweighing only	α = 0.05 (one-sided), power = 0.80, margin δ = 0.01, r = 0.70 (sensitivity 0.50 to 0.85), A = 0.78	14,086 (23,474 at r = 0.50)

Note. All calculations use the dataset's observed prevalence (22.12%) and sex split (39.63% male). AUC = area under the receiver operating characteristic curve. FDR = false-decline rate. Threshold adjustment in RQ4 is evaluated with bootstrap intervals on decision-level measures and is not sized by power.

Final Sample Size Decision

The final minimum sample size is the maximum across the four research questions under the primary assumptions: $N_{min} = \max(14{,}587, 6{,}329, 2{,}314, 14{,}086) = 14{,}587$, set by the AUC precision target in RQ1. The dataset holds 30,000 cardholders, 2.06 times the minimum, and I

will use the full file for every research question rather than subsample. The most conservative sensitivity case, RQ4 at $r = 0.50$, requires 23,474, which the full file also covers. If a compute constraint ever forced a reduction, 14,587 is the floor, drawn by stratified sampling on sex and default status with seed 640 so that the Table 1 allocation ratios hold.

Data Description

Data Source

The study uses dataset 350, Default of Credit Card Clients, from the UCI Machine Learning Repository, derived from the study by Yeh and Lien (2009) and donated to the repository on 25 January 2016. The file describes 30,000 cardholders of a Taiwanese card issuer observed from April to September 2005, with the target recording whether each cardholder missed the October 2005 payment. It is available at https://archive.ics.uci.edu/dataset/350/default+of+credit+card+clients under the persistent identifier https://doi.org/10.24432/C55S3H, and the repository's recommended citation is Yeh (2009). The repository distributes one Excel file of 5.3 MB with 30,000 rows and 25 columns: a row identifier, 23 predictors, and the binary target, which equals 1 for 6,636 cardholders (22.12%), under a Creative Commons Attribution 4.0 licence with no missing values reported.

Data Dictionary

Table 2 defines every variable in the file, the four engineered features, and the two simulated decision variables. Where the repository documentation is silent, the table records my interpretation as an assumption.

Table 2
Data Dictionary for the Default of Credit Card Clients Dataset

Variable	Type	Meaning	Units or coding	Role in study
ID	Integer	Row identifier assigned by the data owner	1 to 30,000	Excluded from modelling; keys the out-of-fold predictions

Variable	Type	Meaning	Units or coding	Role in study
LIMIT_BAL	Numeric, continuous	Credit limit granted, including supplementary cards	NT$, 10,000 to 1,000,000	Independent; candidate proxy for sex
SEX	Categorical, nominal	Sex as recorded by the issuer in 2005; two administrative categories only	1 = male, 2 = female	Primary protected attribute (RQ2 to RQ4); excluded from the unaware model
EDUCATION	Categorical, ordinal	Highest education attained	1 = graduate school, 2 = university, 3 = high school, 4 = other; values 0, 5, 6 are undocumented and kept as one distinct level (assumption)	Independent; excluded from the unaware model
MARRIAGE	Categorical, nominal	Marital status	1 = married, 2 = single, 3 = other; value 0 is undocumented and kept as a distinct level (assumption)	Independent; excluded from the unaware model
AGE	Numeric, discrete	Age at observation	Years, 21 to 79	Independent; source of AGE_BAND
PAY_0, PAY_2 to PAY_6	Categorical, ordinal (6 fields)	Repayment status in September, August, July, June, May, and April 2005	1 to 9 = months of payment delay (documented); −2 and 0 are undocumented, interpreted as no consumption and revolving use (assumption, tested by sensitivity)	Independent; behavioural signal, strongest expected predictors
BILL_AMT1 to BILL_AMT6	Numeric, continuous (6 fields)	Statement balance, September through April 2005	NT$; negative values are credit balances	Independent
PAY_AMT1 to PAY_AMT6	Numeric, continuous (6 fields)	Amount paid, September through April 2005	NT$, non-negative	Independent

Variable	Type	Meaning	Units or coding	Role in study
DEFAULT (default payment next month)	Binary	Observed missed payment in October 2005	1 = default, 0 = no default	Dependent variable for RQ1 to RQ4
AGE_BAND (derived)	Categorical, ordinal	Age grouped into four bands	< 30, 30 to 39, 40 to 49, ≥ 50	Secondary protected attribute (RQ2)
UTILISATION (derived)	Numeric, continuous	Latest statement balance divided by credit limit	Ratio, BILL_AMT1 / LIMIT_BAL; negative where the balance is a credit, above 1 where over limit; both retained	Independent, engineered
PAY_RATIO (derived)	Numeric, continuous	Latest payment divided by prior statement balance	Ratio, PAY_AMT1 / BILL_AMT2, capped at 1; set to 1 where BILL_AMT2 ≤ 0 (nothing owed)	Independent, engineered
MAX_DELAY (derived)	Numeric, discrete	Worst repayment-status code over six months	Code, −2 to 9, max of PAY_0 to PAY_6; not a duration	Independent, engineered
FLAG (derived decision)	Binary	Simulated referral or decline: score above the cut-off	1 = flagged, 0 = not flagged	Numerator of the false-decline rate within non-defaulters (RQ3, RQ4)
RETAIN (derived decision)	Binary	Simulated favourable decision: 1 − FLAG	1 = retained, 0 = flagged	Favourable-selection rate by group; numerator and denominator of the DI ratio (RQ3, RQ4)

Note. NT$ = New Taiwan dollars. DI = disparate impact. The unaware model is the configuration audited in RQ3, trained with SEX, EDUCATION, and MARRIAGE removed. An aware model including them is trained for comparison. Variable names follow the column headers in the source file.

Known Quality Issues and Cleaning Plan

Three coding problems are visible in the raw file. EDUCATION contains values 0, 5, and 6 that the documentation does not define, in roughly 345 rows, and MARRIAGE contains an undocumented 0 in 54 rows. Both are kept as a distinct undocumented level rather than merged

into the documented other category, so that the assumption stays visible, and a sensitivity run merges them to confirm the choice does not move the results. The repayment-status fields contain −2 and 0, which the repository does not define. I interpret them as no consumption and revolving use, as published analyses commonly do, and a sensitivity run collapses both into one non-delinquent code. The identifier is unique, so no de-duplication is required. Monetary variables are right-skewed and are log-transformed for the logistic model only, with the transform fitted inside each training fold. No row is dropped at any step, so the evaluation pool stays at 30,000.

Data Access and Repository

All code, the data dictionary, and every result table and figure will be held in a public GitHub repository at https://github.com/mriyer-walsh/qm640-credit-fairness, whose planned layout and run instructions appear in Appendix A. At the time of this synopsis the repository holds the README with the dataset citation and licence, the data dictionary as a CSV, the download script that fetches the file and verifies its SHA-256 checksum, and the sample-size script that reproduces Table 1. The notebooks and results folders fill as each week in Table 4 completes, and Appendix A marks which items exist and which are planned. Every random process uses the seed 640.

Ethical Considerations and Limitations

The file contains no names, account numbers, addresses, or dates of birth, and the identifier is a sequential index with no link to the issuer's systems. The data were released for secondary analysis under an open licence, and I will make no attempt at re-identification or linkage with any other source. Sex and marital status are present because the issuer collected them in 2005, and SEX holds two administrative categories that do not describe the range of gender identities. I use these fields only to measure disparity, never as inputs to the audited model, and the final report will say so. A measured disparity is evidence of a group-level difference in simulated decisions, not a finding of unlawful discrimination, which depends on law, intent, and business justification that this study does not examine. Passing the two

benchmarks does not establish regulatory compliance anywhere: the four-fifths heuristic comes from United States employment-selection guidelines, not Indian credit regulation, and the ±0.05 tolerance is a research choice.

Bias in the data has three sources the final report will carry as limitations. The sample over-represents women (60%) and university-educated cardholders (47%), so group-level findings describe this issuer's 2005 customer base. The default label is one issuer's collection outcome in one month, so a model that learns it inherits any unevenness in how the issuer pursued late payers. The file holds no income, bureau score, or regional field, and the attributes an Indian lender would audit first, such as religion, caste, or state, are absent. The study transfers to Indian retail lending as an audit method and a decision table, not as a finding about Indian borrowers.

Analytic Approach

Table 3 maps each research question to its hypothesis, method, assumptions, and outputs. The subsections that follow justify each method and give the evaluation metrics, the pipeline (Figure 1), and the ten-week plan (Table 4).

Table 3
Mapping of Research Questions to Hypotheses, Methods, Assumptions, and Outputs

RQ	Hypothesis tested	Analysis method	Key assumptions	Outputs
RQ1	H1: AUC of gradient boosting exceeds AUC of logistic regression (pre-specified)	Logistic regression, random forest, XGBoost tuned by inner CV on the fit partition; DeLong test on held-out folds; calibration check	Rows independent and identically distributed; logistic assumes linearity in the logit; trees need adequate leaf counts	AUC with 95% CI per model, DeLong p value, KS, Brier, reliability curve, ROC plot
RQ2	H2: default rate differs by sex; H2b: default and age band are not independent	Two-proportion z test with 95% CI; chi-square test with Cramér's V; logistic regression of DEFAULT on SEX adjusting for AGE and LIMIT_BAL as a descriptive sensitivity check	Independent observations; expected cell counts ≥ 5; normal approximation valid ($np \geq 10$)	Group rates, difference with CI and p value, Cramér's V, adjusted odds ratio with CI

RQ	Hypothesis tested	Analysis method	Key assumptions	Outputs
RQ3	H3: false-decline rate differs by sex at 75% simulated retention	Cut-off set on calibration partition; per-sex confusion matrices on held-out folds; RETAIN rate, DI ratio, FDR gap, TPR gap, equalized odds; 1,000-sample stratified bootstrap CIs; aware vs unaware comparison	75% retention is a stated scenario, not an observed lender policy; held-out decisions exchangeable across folds	Fairness table by group with CIs; descriptive comparison with the 0.80 and ±0.05 benchmarks; candidate proxy features (optional SHAP)
RQ4	H4: reweighed AUC non-inferior within 0.01; threshold adjustment judged on decision-level measures	Reweighing (pre-processing) with DeLong non-inferiority test; group thresholds found by direct search on the calibration partition at common 75% retention; bootstrap CIs; trade-off table across retention rates 65% to 85%	Correlation between paired AUCs ≥ 0.70 (sensitivity 0.50 to 0.85); weights and thresholds set on calibration data only	FDR gap and DI before and after each method; AUC loss with CI (reweighing); default rate among retained at 75% with CI (both); decision table

Note. CV = cross-validation. DI = disparate impact ratio. FDR = false-decline rate. KS = Kolmogorov–Smirnov statistic. SHAP = Shapley additive explanations. TPR = true-positive rate.

Solution to RQ1

RQ1 is answered with three supervised classifiers trained on identical fit partitions and scored on identical held-out folds. Logistic regression with L2 regularisation, standardised numeric inputs, and one-hot encoded categories is the baseline, because it is the model most Indian lenders still hold in production and the one a validation unit can read line by line. Random forest (Breiman, 2001) with 500 trees is a secondary benchmark. XGBoost (Chen & Guestrin, 2016) is the challenger, with learning rate, depth, subsampling, and minimum child weight tuned by randomised search inside the fit partition so that no held-out data influences a tuning choice. The contrast is fixed in advance as XGBoost against logistic regression, so the DeLong test is not applied to whichever model happens to score highest. Class imbalance is handled by class weighting. Because weighting can distort calibration, the Brier score is paired with a reliability curve and, if needed, isotonic recalibration on the calibration partition before the probabilities set any cut-off. Lessmann et al. (2015) showed that these three families bracket the achievable range

on credit data, which is why I add no further algorithms, although their benchmark predates XGBoost itself. The outputs are the AUC with a 95% CI for each model, the DeLong *p* value for the pre-specified contrast, the KS statistic, the Brier score, a reliability curve, and a ROC plot.

Solution to RQ2

RQ2 is a statistical question. The sex comparison uses a two-sided two-proportion *z* test with a pooled standard error and a 95% CI for the difference, the test the RQ2 power calculation was built for. The age-band comparison uses a chi-square test of independence on the 2 × 4 table with Cramér's *V* as the effect size, so that a significant result on 30,000 rows is not mistaken for a material one. A logistic regression of DEFAULT on SEX with AGE and LIMIT_BAL as covariates then shows how much of the raw difference survives adjustment, reported as an adjusted odds ratio with a CI and read as a descriptive check rather than a causal estimate. The assumptions, independent observations, expected cell counts of at least five, and samples large enough for the normal approximation, hold at this scale. If base rates differ by sex, equal error rates rather than demographic parity lead in RQ3 and RQ4 (Kleinberg et al., 2017).

Solution to RQ3

RQ3 audits the simulated decisions of the XGBoost model trained without SEX, EDUCATION, and MARRIAGE. On the calibration partition of each outer fold, the score quantile that retains 75% of cardholders is found, and that cut-off is applied to the held-out fold, giving FLAG and RETAIN for every cardholder from a cut-off they did not influence. The 75% retention rate is a stated scenario, not an observed policy, and the RQ4 trade-off table varies it. A confusion matrix is computed for each sex on the pooled held-out decisions, from which the RETAIN rate, the false-decline rate among non-defaulters, and the true-positive rate among defaulters are read. The DI ratio is computed on RETAIN rates, the favourable outcome. The false-decline gap is the primary fairness measure because it is the borrower-facing error, a creditworthy cardholder wrongly flagged, and the true-positive gap is the secondary, lender-risk measure. Confidence intervals come from 1,000 bootstrap resamples stratified by sex and outcome, and the false-decline gap is tested by the two-proportion test the RQ3 power

calculation was sized for. The same audit is repeated for the aware model that includes the three demographic fields, so the final report can state how much unawareness changes the result. If time allows, a SHAP decomposition by sex lists the features whose attributions differ most between groups, with LIMIT_BAL the leading candidate, as candidates for investigation rather than proof of a proxy.

Solution to RQ4

RQ4 applies two mitigation methods separately to the audited model and records what each costs on the measure appropriate to it. Reweighing (Kamiran & Calders, 2012) assigns each row of the fit partition a weight $w(s, y) = P(s)P(y)/P(s, y)$, where s is sex and y is the outcome, so that sex and default are independent in the weighted training set. The model is retrained with those weights, the 75% cut-off is recalibrated, and the held-out fold is scored. Because the scores change, the DeLong non-inferiority test with a 0.01 margin applies, alongside the decision-level measures. Threshold adjustment (Hardt et al., 2016) leaves the scores untouched and sets a separate cut-off per sex. On the calibration partition I search for the pair of group thresholds that equalises the false-decline rates while holding overall retention at 75%, which keeps the two configurations comparable. Fairlearn's ThresholdOptimizer (Bird et al., 2020) is run as a cross-check, with the caveat that some of its constraints yield randomised decisions. For threshold adjustment the AUC is unchanged by construction, so its cost is the change in the default rate among retained cardholders at 75% retention, with a bootstrap CI. A trade-off table then repeats the audit at retention rates from 65% to 85% in five-point steps for the unmitigated and both mitigated configurations, giving retention rate, default rate among retained, AUC, DI ratio, and false-decline gap for each row. Group-specific thresholds require the protected attribute at decision time and constitute disparate treatment in some jurisdictions, so the table presents reweighing as the single-threshold option, quantifies the extra cost of that constraint, and leaves the choice to the lender's legal review rather than recommending either method.

Evaluation Metrics

The classification metrics are defined from the confusion matrix of true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN), where a positive is a predicted default, so FLAG = 1 is the positive prediction and RETAIN = 1 − FLAG.

$$Precision = \frac{TP}{TP + FP}, TPR = \frac{TP}{TP + FN}, FDR = FPR = \frac{FP}{FP + TN} \tag{3}$$

$$F_1 = \frac{2 \times Precision \times TPR}{Precision + TPR} \tag{4}$$

$$AUC = \frac{1}{n_D n_N} \Sigma_{i,j} I_{ij} \tag{5}$$

$$Brier = \frac{1}{N} \sum_{i=1}^{N} \left(\hat{p}_i - y_i \right)^2 \tag{6}$$

In Equation 3 the false-decline rate (FDR) is the false-positive rate read from the borrower's side: the share of non-defaulters who are flagged. In Equation 5, I_{ij} equals 1 when defaulter *i* scores above non-defaulter *j* and 0 otherwise, so the AUC is the probability that a random defaulter outranks a random non-defaulter. It does not depend on the cut-off. The KS statistic is KS = max |TPR(*t*) − FPR(*t*)| over thresholds *t*. The Brier score in Equation 6 measures calibration as the mean squared gap between predicted probability $\hat{p}_i$ and outcome y_i. The fairness measures compare the two sex groups *g*. The favourable-selection rate is $\text{RETAIN}_g = (\text{TN}_g + \text{FN}_g)/n_g$. The disparate impact ratio is DI = min(RETAIN$_g$)/max(RETAIN$_g$), and the four-fifths benchmark is met when DI ≥ 0.80 (Feldman et al., 2015). The false-decline gap is $\text{FDR}_{\text{male}} - \text{FDR}_{\text{female}}$, which is the equal opportunity difference of Hardt et al. (2016) once the favourable outcome is taken as non-default. The true-positive gap $\text{TPR}_{\text{male}} - \text{TPR}_{\text{female}}$ is the lender-risk counterpart, and the equalized odds difference is the larger of the two absolute gaps. The default rate among retained cardholders, total FN divided by total RETAIN, is the simulated portfolio outcome at each cut-off.

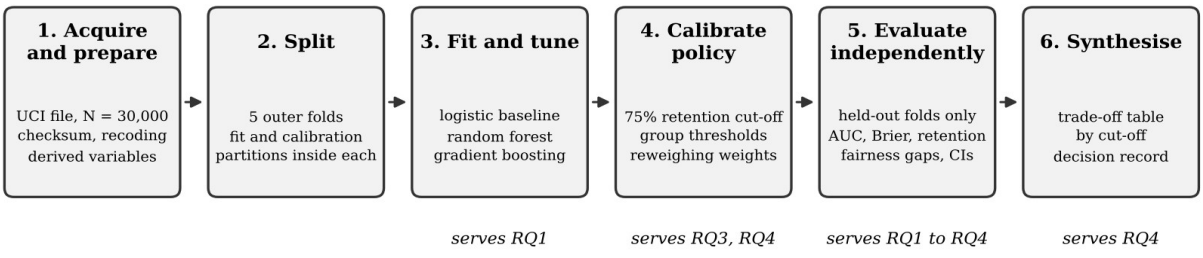
A worked calculation shows how the audit reads. Suppose the male group holds 1,000 cardholders of whom 240 default, the cut-off flags 280, and 132 of the flagged are true defaulters. Then $RETAIN_{male}$ = 720/1,000 = 0.720, FDR_{male} = 148/760 = 0.195, TPR_{male} = 132/240 = 0.550, and the default rate among retained men is 108/720 = 0.150. Suppose the female group holds 1,500 cardholders of whom 312 default, 330 are flagged, and 156 of those are true defaulters. Then $RETAIN_{female}$ = 1,170/1,500 = 0.780, FDR_{female} = 174/1,188 = 0.146, TPR_{female} = 156/312 = 0.500, and the default rate among retained women is 156/1,170 = 0.133. The audit gives DI = 0.720/0.780 = 0.923, which meets the 0.80 benchmark, a false-decline gap of 0.195 − 0.146 = 0.049, inside the ±0.05 band but at its edge, a true-positive gap of 0.050, and an equalized odds difference of 0.050. Had flag rates been used in place of RETAIN rates, the ratio would have read 0.220/0.280 = 0.786 and the benchmark would appear failed, which is why the favourable outcome is fixed before any metric is computed. A lender reading this table learns that creditworthy men are wrongly flagged about five points more often than creditworthy women at this cut-off, which is where the RQ4 trade-off table becomes the decision document.

Solution Pipeline

Figure 1 shows the six-stage pipeline from data acquisition to the decision table. Every stage is a script or notebook in the repository, and the pipeline keeps model fitting, policy calibration, and evaluation on disjoint data within each outer fold.

Figure 1

Solution Pipeline From Data Acquisition to Decision Table



Stage 4 uses only the calibration partition of each outer training set. Stage 5 scores only the held-out fold. Every run is seeded.

Note. Stages 1 and 2 run once and feed all four research questions. Stage 3 answers RQ1, stage 4 sets the cut-offs and weights for RQ3 and RQ4, stage 5 produces every reported estimate, and stage 6 assembles the RQ4 trade-off table. All code is in the repository described in Appendix A.

Stages 1 and 2 prepare the data and fix the fold and partition assignment, keyed on ID and stratified by sex and outcome. Stages 3 and 4 fit the models on the fit partition and set every cut-off and weight on the calibration partition. Stage 5 scores each held-out fold once and computes every reported estimate with its interval, and stage 6 assembles the trade-off table and the decision record into one results workbook committed to the repository.

Project Plan

Table 4 sets out the ten-week plan. The first two weeks are front-loaded with data work because every later week depends on a clean, fold-assigned file. RQ2 precedes RQ1 because its result decides which fairness criterion leads in RQ3 and RQ4. The interim report falls at the end of week 6, after RQ1 to RQ3 are complete.

Table 4
Ten-Week Project Plan

Week	Phase	Activities	Deliverable
1	Set-up and acquisition	Create repository and environment; download and checksum the file; verify data dictionary against the source; agree scope and benchmarks with mentor	Repository with README, dictionary, download and sample-size scripts; verified raw file
2	EDA and splitting	Univariate, bivariate, and multivariate EDA; recoding; derived variables; group tabulations by sex and age band; outer folds and fit/calibration partitions; runtime benchmark	EDA notebook, cleaned Parquet file, fold file
3	RQ2 statistical tests	Two-proportion test, chi-square with Cramér's V, adjusted logistic regression	RQ2 results table with CIs
4	Baseline modelling	Logistic regression pipeline; nested CV framework; calibration check; KS, Brier; held-out scoring	Baseline AUC and CI
5	Challenger modelling	Random forest and XGBoost with randomised search; DeLong test on the pre-specified contrast; aware and unaware runs	RQ1 results, ROC and reliability figures

Week	Phase	Activities	Deliverable
6	Fairness audit	Cut-off on calibration partitions; per-sex confusion matrices on held-out folds; DI, FDR gap, TPR gap, equalized odds; bootstrap CIs	RQ3 fairness table; interim report submitted
7	Mitigation, reweighing	Compute weights on calibration partitions; retrain on fit partitions; non-inferiority test; fairness re-audit	RQ4a results
8	Mitigation, threshold adjustment	Group thresholds by direct search at common retention; ThresholdOptimizer cross-check; trade-off table across retention rates; optional SHAP	RQ4b results, trade-off table
9	Synthesis	Decision record, recommendations, limitations, next steps; code freeze; README and run instructions finalised	Draft final report, frozen repository
10	Final report	Mentor review, revisions, presentation, submission	Final report and presentation

Note. EDA = exploratory data analysis. Each week assumes roughly twelve hours of effort alongside full-time employment.

Recommendation and Application

The primary users are the model validation unit and the chief risk officer of a retail lender, because they sign the model into production and answer for it to the supervisor. The secondary users are the head of retail credit, who owns the cut-off, and the compliance function that prepares the AI disclosures the FREE-AI report recommends (Reserve Bank of India, 2025). A third audience is the mid-sized NBFC that buys scores from a vendor and cannot demand the model's internals. The audit needs only scores, outcomes, and the protected attribute joined at evaluation time, so it works as an independent check on a vendor model, the position most of my NBFC clients are in. In every case the output is a decision-support table, and a lender would still need its own labels, costs, risk appetite, and legal review before acting on it.

The first recommendation the study will support is to replace the single-threshold sign-off in model validation with the RQ4 trade-off table. For each candidate cut-off the table shows the simulated retention rate, the default rate among retained customers, the AUC, the DI ratio, and the false-decline gap, and the credit policy committee chooses a row and signs that row. This

turns a fairness policy into a configuration with a named owner and a dated approval, which a supervisor can audit. The second is to set a fairness tolerance, for instance DI of at least 0.80 and an absolute false-decline gap of at most 0.05, as a hard gate that cannot be traded against bad rate once set, and to record the cost of meeting it in the model inventory. The two values used here are research benchmarks, and a lender would set its own. The third is to audit the production feature set rather than the policy: the RQ3 result will show how much excluding sex changes the simulated decisions, and I expect the answer to be little, because credit limit and repayment history carry most of the same information. The fourth concerns method choice: where the law treats group-specific cut-offs as disparate treatment, reweighing with a single threshold is the usable option, and RQ4 will state what that constraint costs. The fifth is operational: a validation analyst at an Indian issuer can clone the repository, point it at the issuer's own extract with the same column names, and run the audit on a retrospective portfolio, with the understanding that the method transfers and the Taiwan results do not. The final report will deliver the fitted results for RQ1 to RQ4, the trade-off table, the decision record, a limitations section, and the repository.

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Appendix A

Repository Layout and Run Instructions

The public repository at https://github.com/mriyer-walsh/qm640-credit-fairness holds the code, data dictionary, and results as they are produced. The folder tree below is the planned layout from the repository README. Items marked (at synopsis) exist at submission, and the remainder are created in the week shown in Table 4.

```
qm640-credit-fairness/
|-- README.md                 (at synopsis) summary, licence, citation
|-- requirements.txt          (at synopsis) pinned dependencies
|-- Makefile                  make data | make all | make report
|-- data/
|   |-- raw/                  filled by download.py, not committed
|   |-- processed/
|   |   |-- clients_clean.parquet      week 2
|   |   |-- folds.csv         week 2: ID, fold, partition, seed 640
|   |-- data_dictionary.csv   (at synopsis) source of Table 2
|-- notebooks/
|   |-- 01_eda.ipynb                   week 2
|   |-- 02_rq2_outcome_disparity.ipynb week 3
|   |-- 03_rq1_models.ipynb            weeks 4 to 5
|   |-- 04_rq3_fairness_audit.ipynb    week 6
|   |-- 05_rq4_mitigation.ipynb        weeks 7 to 8
|-- src/
|   |-- download.py           (at synopsis) fetch, verify checksum
|   |-- sample_size.py        (at synopsis) reproduces Table 1
|   |-- prepare.py            week 2: recoding, features, folds
|   |-- train.py              weeks 4 to 5: models, nested CV
|   |-- policy.py             weeks 6 to 8: cut-offs, thresholds
|   |-- evaluate.py           week 6: metrics, DeLong, bootstrap
|-- results/
|   |-- tables/               one CSV per research question
|   |-- figures/              ROC, reliability, trade-off curve
|-- reports/
    |-- synopsis.pdf          (at synopsis)
    |-- interim_report.pdf    week 6
    |-- final_report.pdf      week 10
```

To reproduce the analysis once the pipeline is complete, clone the repository, create a Python 3.11 virtual environment, and install the pinned dependencies with pip install -r requirements.txt. The command make data runs download.py and prepare.py, which fetch the source file from the UCI repository, verify it against the stored checksum, apply the cleaning plan, and write the Parquet file and fold assignment. The command make all executes the five

notebooks in order and writes every table and figure to the results folder. The command make report rebuilds the PDF reports from those outputs. Running sample_size.py alone prints the calculations behind Table 1 and works at submission. Runtime will be recorded in the README after the week 2 benchmark.